HDOC 16

Menu-Driven Digit Operations Using Switch-Case

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Aim / Question

Write a menu-driven C program using **switch-case** for a positive integer. Use one separate **void function** for every menu option. Each function takes the input itself, processes it, and displays the result.

1. Check whether digits are in increasing order from left to right.
2. Check whether digits are in decreasing order from left to right.
3. Find the largest and smallest digit.
4. Find the second largest and second smallest digit.

Algorithm

1. Start the program.
2. Display four menu options.
3. Read the user choice.
4. Use switch-case to call the required function.
5. Inside that function, read the positive integer.
6. Perform the selected digit operation and print the answer.
7. Stop.

Test Case Table

Choice	Input	Expected Result
1	123579	Digits are in increasing order
1	12432	Digits are not in increasing order
2	975310	Digits are in decreasing order
3	58219	Largest = 9, Smallest = 1
4	58317	Second largest = 7, Second smallest = 3

C Program

```
#include <stdio.h>

void increasing();
void decreasing();
void largestSmallest();
void secondLargestSmallest();

int main() {
    int choice;

    printf("1. Increasing order\n");
    printf("2. Decreasing order\n");
    printf("3. Largest and smallest digit\n");
    printf("4. Second largest and second smallest digit\n");
    printf("Enter choice: ");
    scanf("%d", &choice);

    switch(choice) {
        case 1: increasing(); break;
        case 2: decreasing(); break;
        case 3: largestSmallest(); break;
        case 4: secondLargestSmallest(); break;
        default: printf("Invalid choice\n");
    }
    return 0;
}

void increasing() {
    int n, a[20], i = 0, j, ok = 1;
    printf("Enter number: ");
    scanf("%d", &n);

    while(n > 0) {
        a[i++] = n % 10;
        n /= 10;
    }
    for(j = i - 1; j > 0; j--)
        if(a[j] >= a[j-1]) ok = 0;

    if(ok) printf("Digits are in increasing order\n");
    else printf("Digits are not in increasing order\n");
}

void decreasing() {
    int n, a[20], i = 0, j, ok = 1;
    printf("Enter number: ");
    scanf("%d", &n);

    while(n > 0) {
        a[i++] = n % 10;
        n /= 10;
    }
    for(j = i - 1; j > 0; j--)
        if(a[j] <= a[j-1]) ok = 0;

    if(ok) printf("Digits are in decreasing order\n");
    else printf("Digits are not in decreasing order\n");
}

void largestSmallest() {
    int n, d, largest = 0, smallest = 9;
    printf("Enter number: ");
    scanf("%d", &n);

    while(n > 0) {
        d = n % 10;
        if(d > largest) largest = d;
        if(d < smallest) smallest = d;
        n /= 10;
    }
    printf("Largest = %d\nSmallest = %d\n", largest, smallest);
}

void secondLargestSmallest() {
    int n, d, max1=-1, max2=-1, min1=10, min2=10;
    printf("Enter number: ");
    scanf("%d", &n);
```

```
    while(n > 0) {
        d = n % 10;

        if(d > max1) { max2 = max1; max1 = d; }
        else if(d > max2 && d != max1) max2 = d;

        if(d < min1) { min2 = min1; min1 = d; }
        else if(d < min2 && d != min1) min2 = d;

        n /= 10;
    }

    if(max2 == -1 || min2 == 10)
        printf("Second largest/smallest not possible\n");
    else {
        printf("Second largest = %d\n", max2);
        printf("Second smallest = %d\n", min2);
    }
}
```

Compilation and Execution

```
gcc hdoc16.c -o hdoc16
./hdoc16
```

Sample Outputs

```
Choice 1
Enter number: 123579
Digits are in increasing order
```

```
Choice 2
Enter number: 975310
Digits are in decreasing order
```

```
Choice 3
Enter number: 58219
Largest = 9
Smallest = 1
```

```
Choice 4
Enter number: 58317
Second largest = 7
Second smallest = 3
```

Result

The menu-driven program was written using switch-case and separate void functions. The program was compiled and tested with the prepared test cases.